

OWOSSO, MI, USA

WMO: 722346

Lat: 42.993N

Lon: 84.139W

Elev: 224

StdP: 98.66

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.9	-22.1	0.5	-17.2	-18.9	0.7	-13.8	11.3	-5.3	9.8	-3.1	2.0	250	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.2	22.0	30.6	21.6	28.9	20.7	24.1	28.8	23.2	27.7	22.3	26.7	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	26.3	22.0	17.1	25.6	21.1	16.2	24.7	73.7	28.8	69.8	27.4	66.6	26.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.3	-21.1	34.3	3.3	1.4	-23.5	35.3	-25.4	36.1	-27.3	36.9	-29.7	37.9		
			-21.5	25.7	3.2	1.0	-23.8	26.4	-25.7	27.0	-27.5	27.5	-29.8	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-4.2	-3.8	1.7	8.3	14.9	19.9	22.0	20.9	17.4	11.0	4.3	-0.8
	DBStd	10.54	6.03	6.10	6.23	4.97	4.94	3.75	3.25	3.03	4.32	4.90	5.07	4.89
	HDD10.0	1766	442	388	269	91	9	0	0	0	1	47	183	336
	HDD18.3	3633	699	621	517	303	133	24	7	11	69	235	421	593
	CDD10.0	1531	1	1	12	38	161	297	372	338	222	77	12	1
	CDD18.3	357	0	0	1	1	26	71	121	90	41	7	0	0
	CDH23.3	3279	0	0	8	24	276	689	1142	748	334	57	1	0
	CDH26.7	970	0	0	1	2	73	214	382	194	94	10	0	0
Wind	WSAvg	2.9	3.5	3.5	3.3	3.6	3.0	2.6	2.2	2.0	2.2	2.8	3.3	3.4
Precipitation	PrecAvg	753	42	34	50	70	76	82	68	82	85	56	65	52
	PrecMax	1092	108	96	114	146	260	156	157	252	261	152	165	113
	PrecMin	520	6	1	10	14	2	12	11	16	0	10	12	8
	PrecStd	127	26	23	25	32	47	37	34	42	48	30	33	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.4	23.9	26.4	31.2	32.9	34.0	32.6	32.5	28.0	21.0	15.2
		MCWB	11.4	10.7	16.9	15.3	20.8	22.3	23.1	22.2	21.1	20.4	13.6	13.2
	2%	DB	9.4	10.1	17.9	22.8	28.7	31.3	32.2	30.6	29.8	24.9	17.1	10.9
		MCWB	7.4	7.8	12.5	13.3	19.2	21.6	22.6	21.6	21.5	17.6	11.7	9.0
	5%	DB	6.2	6.4	14.1	20.4	26.8	29.1	30.5	28.9	27.2	22.2	14.8	8.0
		MCWB	4.2	4.0	10.3	12.4	17.9	20.5	21.9	21.0	19.5	16.9	10.8	6.4
	10%	DB	3.4	3.9	11.1	17.5	24.0	27.3	28.8	27.4	25.0	19.4	12.3	5.8
		MCWB	2.2	1.9	7.6	11.0	16.5	19.4	20.9	20.4	18.9	15.2	9.1	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.8	17.3	17.6	22.3	24.1	25.7	25.2	24.4	21.5	15.4	13.5
		MCDB	12.4	13.4	23.7	23.1	28.1	30.0	31.2	29.0	28.7	26.0	18.2	14.8
	2%	WB	7.9	7.8	14.0	15.4	20.9	23.0	24.5	23.6	22.8	19.8	13.0	9.3
		MCDB	8.7	10.1	17.2	20.1	25.7	28.8	29.5	27.2	26.4	23.9	16.3	10.4
	5%	WB	4.5	4.3	11.2	13.5	19.7	21.9	23.4	22.6	21.6	17.7	11.0	6.5
		MCDB	6.0	6.1	14.2	19.2	25.0	27.4	28.0	26.4	24.8	20.1	13.8	7.8
	10%	WB	2.4	1.9	7.6	11.7	17.9	20.9	22.4	21.7	20.3	15.6	9.1	4.0
		MCDB	3.5	3.7	10.6	16.7	22.5	25.2	26.8	25.7	23.4	18.8	11.9	5.5
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.0	9.4	11.4	11.7	11.3	11.5	11.2	11.6	10.0	8.1	6.0
		MCDBR	8.9	9.8	13.3	16.0	13.4	12.5	12.7	12.6	13.0	13.9	11.7	9.1
		MCWBR	7.8	8.1	8.8	8.5	6.2	5.4	5.5	5.4	5.9	7.6	7.9	7.7
	5% WB	MCDBR	8.5	9.3	12.6	13.4	11.8	10.5	10.7	10.1	10.4	11.7	10.5	8.7
		MCWBR	7.8	8.2	9.2	8.8	6.3	5.6	5.4	5.2	5.8	8.0	8.3	8.0
Clear-Sky Solar Irradiance	taub		0.294	0.327	0.338	0.379	0.405	0.430	0.417	0.411	0.378	0.344	0.314	0.292
	taud		2.365	2.259	2.368	2.275	2.257	2.219	2.264	2.273	2.357	2.449	2.477	2.425
	Ebn at Noon		840	862	903	886	869	845	852	845	848	839	815	812
	Edh at Noon		80	105	107	128	134	139	132	127	108	87	72	69
All-Sky Solar Radiation	RadAvg		1.45	2.33	3.58	4.58	5.35	6.03	6.10	5.22	4.19	2.66	1.64	1.15
	RadStd		0.13	0.29	0.37	0.41	0.48	0.36	0.29	0.23	0.39	0.26	0.21	0.10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page